

ALAA SHAMMOUT

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SUMMARY

Computer Engineering graduate seeking an 8-week accredited internship in software development, cloud, AI and machine learning, or embedded systems, with substantial project work in each. Full-stack development with React, FastAPI and Spring Boot, plus automation work that a logistics company adopted into daily use. Native Arabic and fluent English. UAE Golden Visa holder, so no employer sponsorship is required. Available to start immediately and full-time for the placement, and open to continuing afterwards.

EDUCATION

American University of Sharjah

Bachelor of Science in Computer Engineering

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Computer Vision

Sharjah, UAE

June 2026

EXPERIENCE

Sarmeen General Transport | Dubai, UAE | *Operations Assistant*

June 2026 – July 2026

- Processed 1,000+ GDRFA entry permits with a single error across the full volume, in a workflow where one incorrect field cost AED 600 to refile.
- Automated permit data entry using an AutoHotkey script for repeated fields, engineered around portal inputs that blocked clipboard paste, plus local on-device OCR to extract passport details instead of retyping them, cutting 2-3 minutes per permit and removing the main source of transcription error.
- Introduced an AI-assisted extraction workflow that converted 20-50 daily paper invoices into formatted Excel records in under a minute, removing manual entry.

PROJECTS

AI-Powered EV Charging Support Platform | *Lead Frontend Developer*

Sep 2025 – May 2026

Industry-sponsored capstone for Emagine Solutions | 4-person team | weekly technical reviews with their software lead

- Led frontend for two React and Tailwind apps, a customer portal and an operator dashboard, integrated with a FastAPI and PostgreSQL backend across 80+ REST endpoints with JWT and OTP authentication.
- Built the operator dashboard on live IoT telemetry: AI-classified tickets with confidence scores, predictive maintenance alerts, and a fleet incident mode triggering on threshold breach.
- Designed a Smart Route feature on the Mapbox Directions API routing drivers to underused chargers via live traffic, with drive/wait/charge estimates and points-based incentives to shift demand off congested stations.

Cloud-Native Booking Microservice | *Java, Spring Boot, PostgreSQL, Azure | 4-person team*

Mar 2026 – May 2026

- Built one of four microservices in Java 21 and Spring Boot: 8 REST endpoints for creation, rescheduling, cancellation and history, with JWT-scoped access and PostgreSQL via Spring Data JPA.
- Enforced integrity with interval-overlap conflict detection, duration and daily limits, and operating-hours checks called over REST against a teammate's resource service; added PostgreSQL advisory locking to reduce double-booking, deployed to Azure App Service and verified endpoints in Postman.

Crowd Counting with Deep Learning | *CSRNet, PyTorch | 3-person team*

Mar 2026 – May 2026

- Re-implemented CSRNet (VGG-16 frontend, dilated backend) in PyTorch on ShanghaiTech with geometry-adaptive k-NN density maps; test MAE 18.01 Part B, 106.11 Part A, ahead of 2 of 4 published baselines on MAE.
- Traced a training failure to density values near $1e-4$ collapsing the MSE loss, fixed with x100 rescaling plus gradient clipping and StepLR decay; per-bucket analysis put sparse Part B scenes at 9.1% relative error.

Multi-Stage Embedded Game on ESP32 | *FreeRTOS, C, ESP-IDF | 3-person team*

Apr 2026 – May 2026

- Built a three-stage interactive escape-room game on ESP32 in C using FreeRTOS, ESP-IDF and PlatformIO, driving a 4x4 keypad, analog joystick, slide potentiometer, RGB LEDs, buzzer and 20x4 I2C LCD across ADC, GPIO, PWM and I2C.
- Structured the system as three periodic tasks with assigned priorities and rates, logic at 20 ms, input at 30 ms and display at 100 ms, coordinating them through a single mutex over shared game state and validating the full state machine in Wokwi.

SKILLS & ACTIVITIES

Programming Languages: Java, Python, JavaScript, C, C++, SQL, Verilog

Backend & Cloud: Spring Boot, Spring Data JPA, FastAPI, REST APIs, PostgreSQL, Microsoft Azure, Maven

Frontend: React.js, Vite, Tailwind CSS, Axios, Mapbox GL, Firebase Cloud Messaging, HTML, CSS, TypeScript

Embedded & ML: FreeRTOS, ESP32, ESP-IDF, PlatformIO, Wokwi, Intel Quartus, PyTorch, OpenCV, NumPy, pandas

Tools: Git, Postman, Linux/Bash, VS Code

Spoken Languages: Arabic (Native), English (Fluent), Turkish (Intermediate)

Activities: Bronze medalist, AUS Men's Swimming (2022) | AI Club, IEEE EMBS, Computer Engineering Club